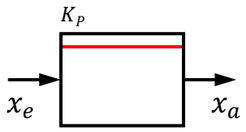
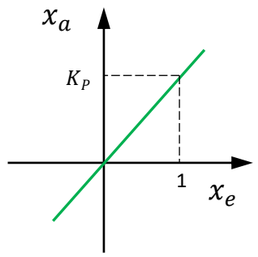
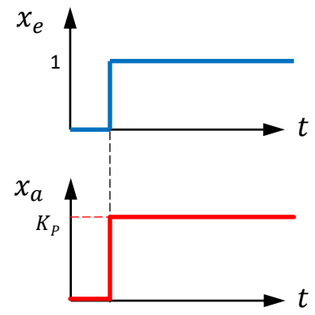
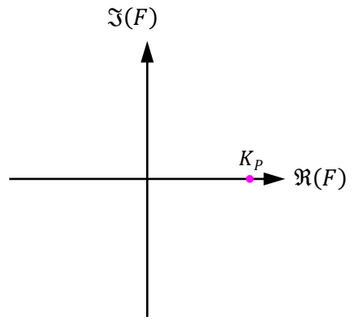
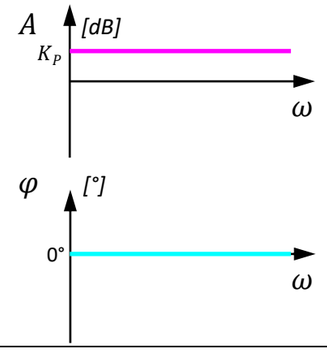
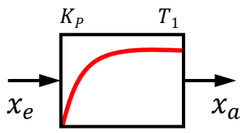
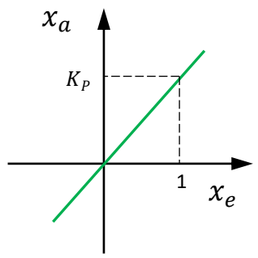
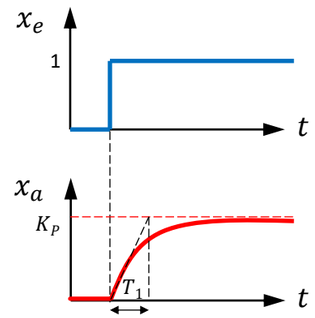
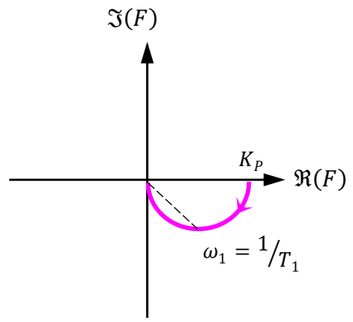
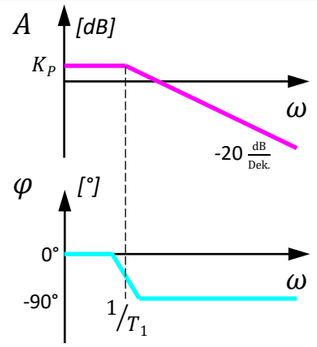
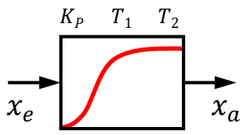
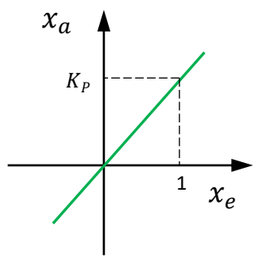
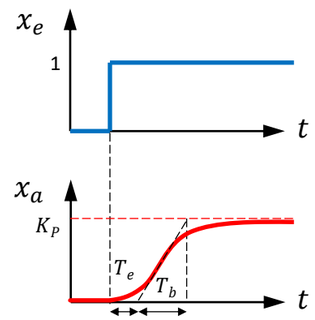
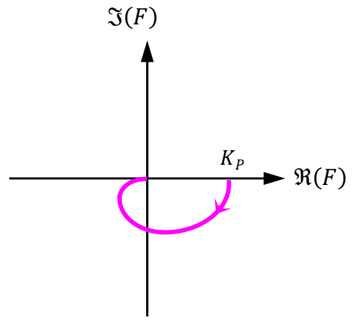
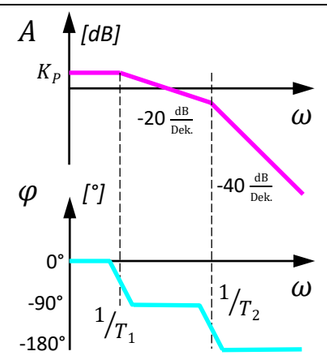
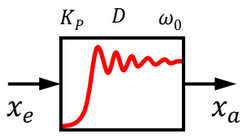
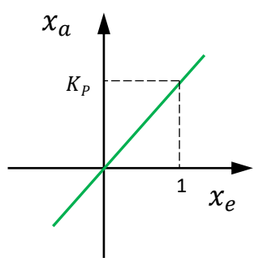
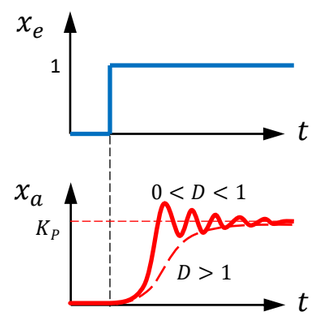
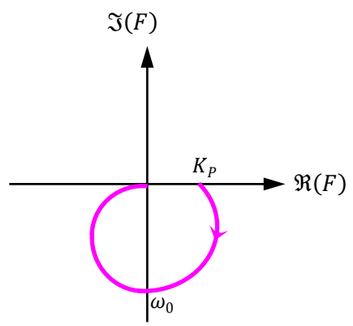
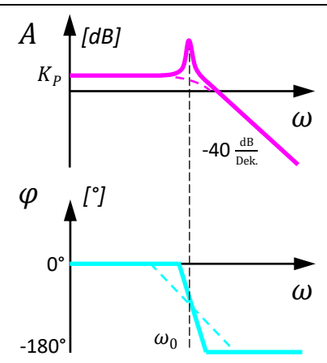
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x_e	Eingang	K_P	Proportionalbeiwert	$T_{1,2,..}$	Zeitkonstante	T_t	Totzeit				
x_a	Ausgang	K_I	Integrierbeiwert (=1/T_I)	T_e	Verzugszeit	D	Dämpfungsgrad	T_i	Nachstellzeit		
t	Zeit	K_D	Differenzierbeiwert (=T_D)	T_b	Ausgleichszeit	ω_0	Kennkreisfrequenz	T_d	Vorhaltezeit		



	x_e	x_a
Allgemein:	x_e	x_a
Regler (RE):	e	y
Regelstrecke (RS):	y	x

Bezeichnung / Funktionsblock	Übertragungsverhalten		Mathematische Funktionen		Grafische Darstellungen	
	statisch	dynamisch	Zeitbereich	Frequenzbereich	Ortskurve	Bodediagramm
P-Glied 			$x_a(t) = K_P \cdot x_e(t)$	$F(j\omega) = K_P$		
P-T_1-Glied 			$T_1 \cdot \dot{x}_a(t) + x_a(t) = K_P \cdot x_e(t)$ Sprungantwort: $x_a(t) = K_P \cdot x_{e0} \cdot \left(1 - e^{\frac{-t}{T_1}}\right)$	$F(j\omega) = \frac{K_P}{1 + j\omega T_1}$ $A(\omega) = \frac{K_P}{\sqrt{1 + (\omega T_1)^2}}$ $\varphi(\omega) = -\tan^{-1}(\omega T_1)$		
P-T_2-Glied 			$T_1T_2 \cdot \ddot{x}_a(t) + (T_1 + T_2) \cdot \dot{x}_a(t) + x_a(t) = K_P \cdot x_e(t)$	$F(j\omega) = \frac{K_P}{(1 + j\omega T_1) \cdot (1 + j\omega T_2)} = \frac{K_P}{1 + j\omega(T_1 + T_2) + (j\omega)^2 T_1 T_2}$		
P-T_2S-Glied 			$\frac{1}{{\omega_0}^2} \cdot \ddot{x}_a(t) + 2\frac{D}{\omega_0} \cdot \dot{x}_a(t) + x_a(t) = K_P \cdot x_e(t)$	$F(j\omega) = \frac{K_P}{1 + j\omega\frac{2D}{\omega_0} + (j\omega)^2 \frac{1}{{\omega_0}^2}}$		

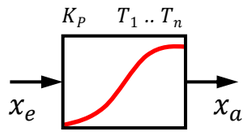
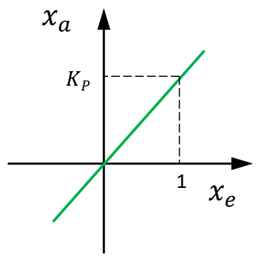
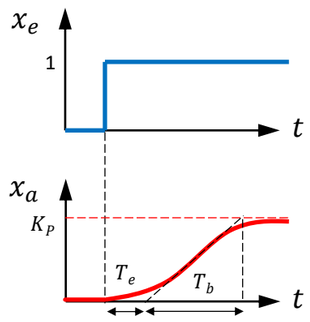
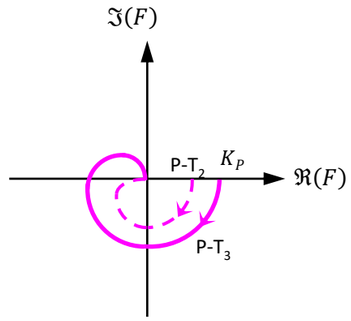
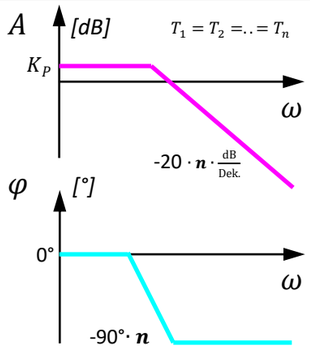
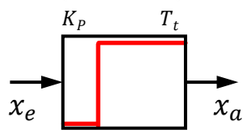
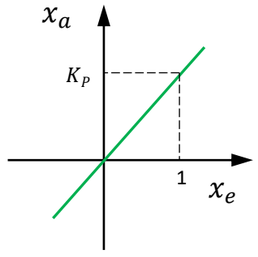
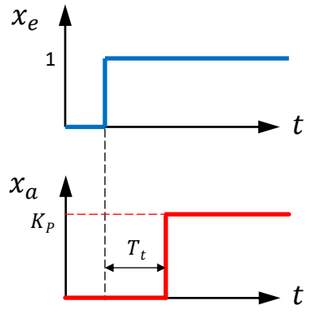
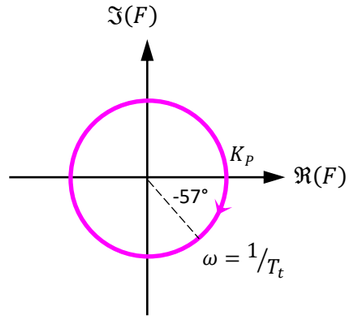
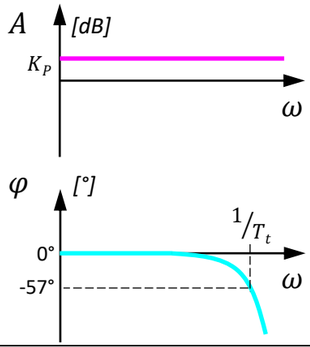
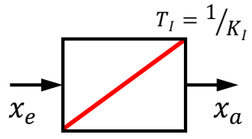
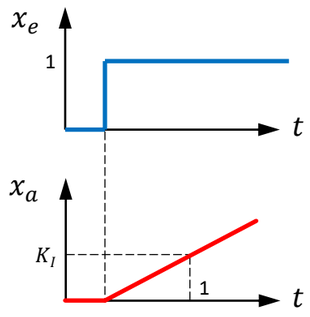
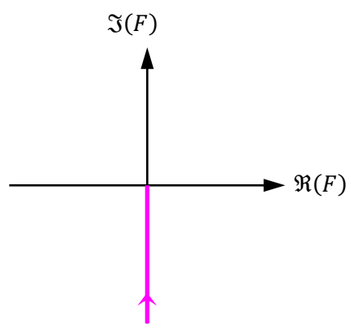
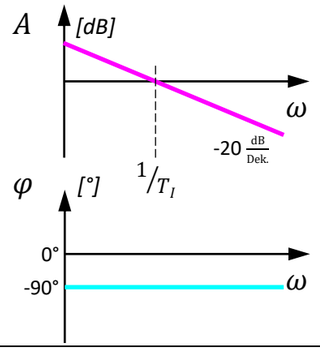
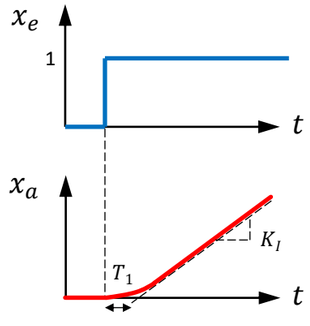
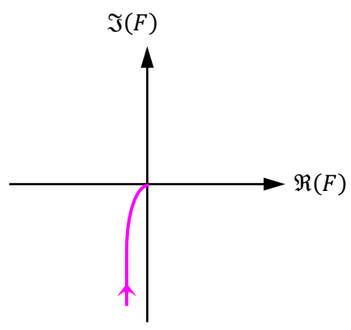
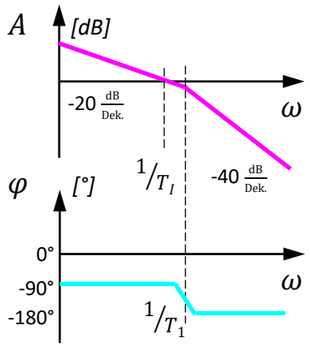
Grundglieder

F	Frequenzgang (komplex)	$\Re(F)$	Realteil	$A = \lvert F \rvert$	Amplitudengang (Betrag) [dB]	$F(j\omega) = \frac{b_0 + j\omega b_1 + (j\omega)^2 b_2 + \cdots + (j\omega)^m b_m}{a_0 + j\omega a_1 + (j\omega)^2 a_2 + \cdots + (j\omega)^n a_n}$	$A_{dB} = 20 \cdot \log_{10}(A)$
ω	Kreisfrequenz [rad/s]	$\Im(F)$	Imaginärteil	$\varphi = \measuredangle F$	Phasengang (Phasenverschiebung) [°]		$A = 10^{\left(A_{dB}/20\right)}$

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x_e	Eingang	K_P	Proportionalbeiwert	$T_{1,2,..}$	Zeitkonstante	T_t	Totzeit				
x_a	Ausgang	K_I	Integrierbeiwert (=1/T_I)	T_e	Verzugszeit	D	Dämpfungsgrad	T_i	Nachstellzeit		
t	Zeit	K_D	Differenzierbeiwert (=T_D)	T_b	Ausgleichszeit	ω_0	Kennkreisfrequenz	T_d	Vorhaltezeit		



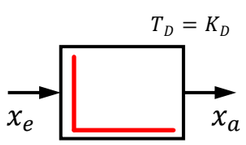
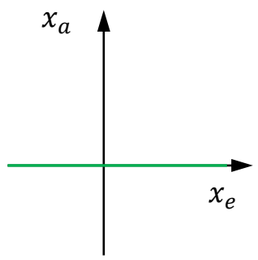
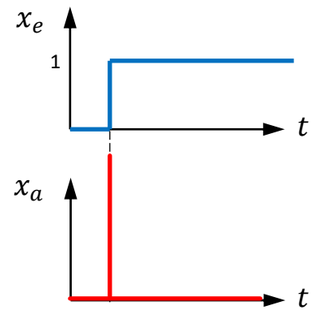
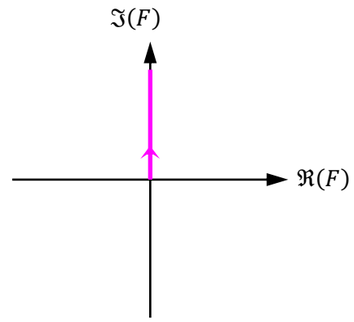
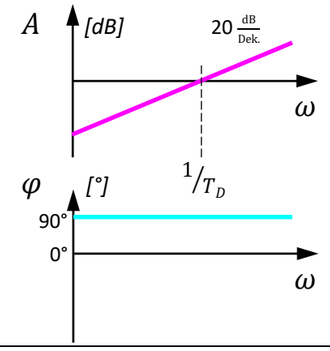
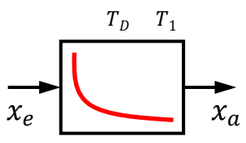
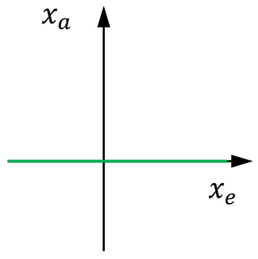
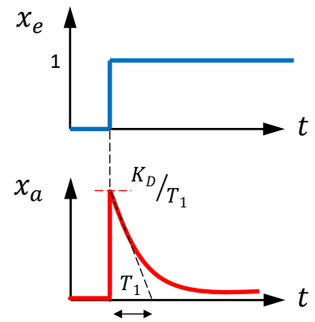
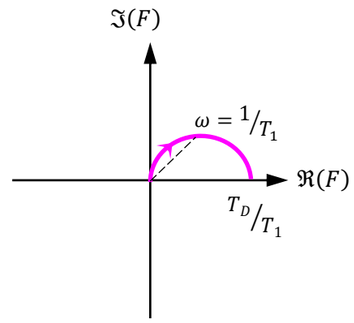
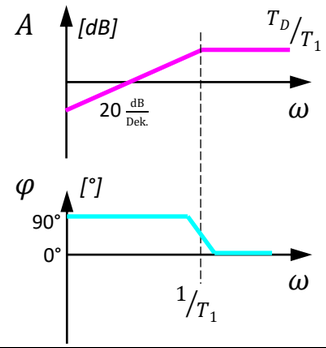
Bezeichnung / Funktionsblock	Übertragungsverhalten		Mathematische Funktionen		Grafische Darstellungen	
	statisch	dynamisch	Zeitbereich	Frequenzbereich	Ortskurve	Bodediagramm
P-T_n-Glied  In BORIS: $T_1 = T_2 = .. = T_n$	 	 	-	$F(j\omega) = \frac{K_P}{(1+j\omega T_1)\cdot \ldots \cdot (1+j\omega T_n)}$ Für $T_1 = T_2 = \cdots = T_n = T$ $F(j\omega) = \frac{K_P}{(1+j\omega T)^n}$	 	
P-T_t-Glied (Totzeit)  	 	 	$x_a(t) = K_P \cdot x_e(t - T_t)$	$F(j\omega) = K_P \cdot e^{-j\omega T_t}$	 	
I-Glied  	-	 	$x_a(t) = K_I \cdot \int x_e(t)\,dt$ $= \frac{1}{T_I} \cdot \int x_e(t)\,dt$	$F(j\omega) = \frac{K_I}{j\omega} = \frac{1}{j\omega T_I}$	 	
I-T_1-Glied Dieses Glied muss in BORIS zusammengesetzt werden aus I-Glied und P-T_1-Glied (analog für I-T_2, I-T_t etc.)	-	 	$T_1 \cdot \dot{x}_a(t) + x_a(t) = K_I \int x_e(t)\,dt$	$F(j\omega) = \frac{K_I}{j\omega \cdot (1+j\omega T_1)} = \frac{1}{j\omega T_1 \cdot (1+j\omega T_1)}$	 	

Grundglieder

F	Frequenzgang (komplex)	$\Re(F)$	Realteil	$A = \lvert F \rvert$	Amplitudengang (Betrag) [dB]	$F(j\omega) = \frac{b_0 + j\omega b_1 + (j\omega)^2 b_2 + \cdots + (j\omega)^m b_m}{a_0 + j\omega a_1 + (j\omega)^2 a_2 + \cdots + (j\omega)^n a_n}$	$A_{dB} = 20 \cdot \log_{10}(A)$
ω	Kreisfrequenz [rad/s]	$\Im(F)$	Imaginärteil	$\varphi = \measuredangle F$	Phasengang (Phasenverschiebung) [°]		$A = 10^{(A_{dB}/20)}$

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x_e	Eingang	K_P	Proportionalbeiwert	$T_{1,2,..}$	Zeitkonstante	T_t	Totzeit						
x_a	Ausgang	K_I	Integrierbeiwert (=1/T_I)	T_e	Verzugszeit	D	Dämpfungsgrad	T_i	Nachstellzeit	Allgemein:	x_e		x_a
t	Zeit	K_D	Differenzierbeiwert (=T_D)	T_b	Ausgleichszeit	ω_0	Kennkreisfrequenz	T_d	Vorhaltezeit	Regler (RE):	e		y
										Regelstrecke (RS):	y		x

Bezeichnung / Funktionsblock	Übertragungsverhalten		Mathematische Funktionen		Grafische Darstellungen	
	statisch	dynamisch	Zeitbereich	Frequenzbereich	Ortskurve	Bodediagramm
D-Glied 			$x_a(t) = K_D \cdot \dot{x}_e(t) = T_D \cdot \dot{x}_e(t)$	$F(j\omega) = j\omega K_D = j\omega T_D$		
D-T_1-Glied 			$T_1 \cdot \dot{x}_a(t) + x_a(t) = K_D \cdot \dot{x}_e(t)$ Sprungantwort: $x_a(t) = \frac{K_D}{T_1} \cdot e^{\frac{-t}{T_1}}$	$F(j\omega) = \frac{j\omega T_D}{1 + j\omega T_1}$		

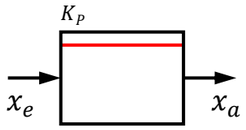
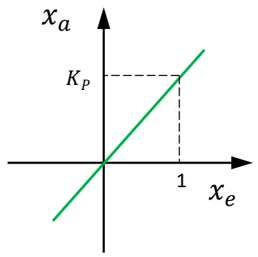
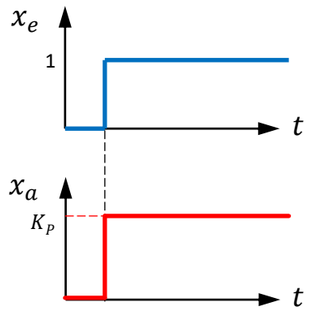
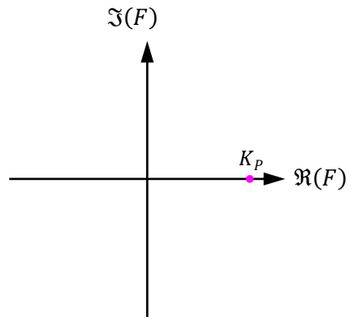
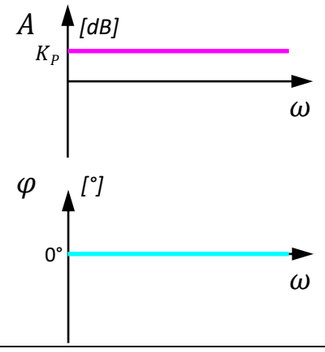
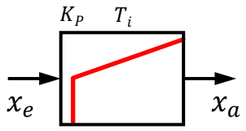
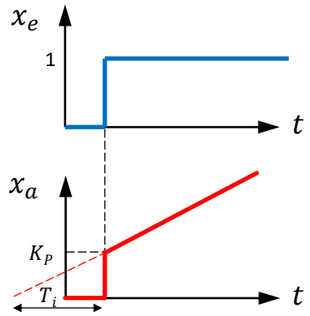
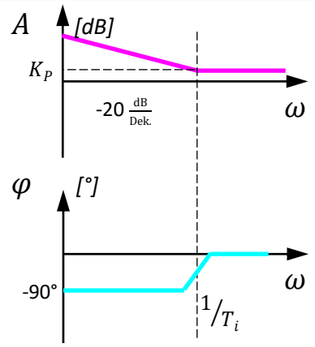
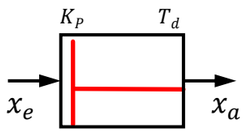
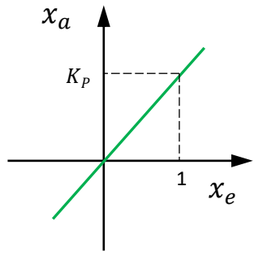
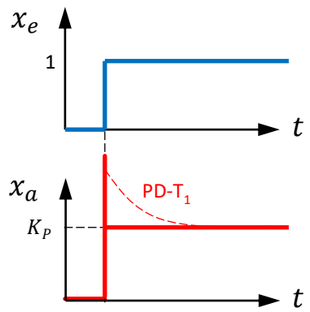
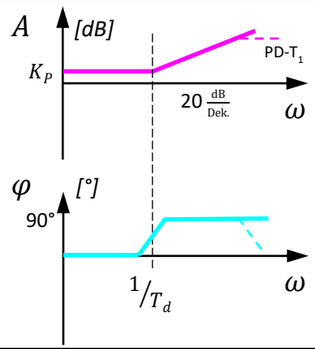
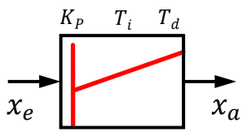
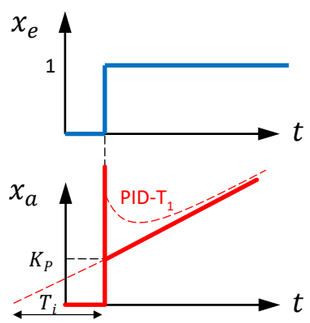
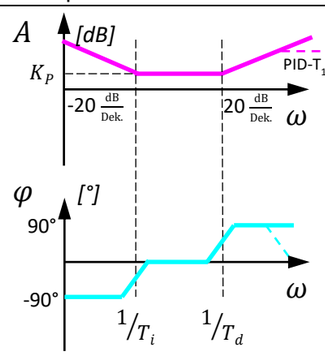
Grundglieder

F	Frequenzgang (komplex)	$\Re(F)$	Realteil	$A = \lvert F \rvert$	Amplitudengang (Betrag) [dB]	$F(j\omega) = \frac{b_0 + j\omega b_1 + (j\omega)^2 b_2 + \cdots + (j\omega)^m b_m}{a_0 + j\omega a_1 + (j\omega)^2 a_2 + \cdots + (j\omega)^n a_n}$	$A_{\mathrm{dB}} = 20 \cdot \log_{10}(A)$
ω	Kreisfrequenz [rad/s]	$\Im(F)$	Imaginärteil	$\varphi = \measuredangle F$	Phasengang (Phasenverschiebung) [°]		$A = 10^{(A_{\mathrm{dB}}/20)}$

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t	Zeit	K_D	Differenzierbeiwert (=T_D)	T_b	Ausgleichszeit	ω_0	Kennkreisfrequenz	T_d	Vorhaltezeit		



Bezeichnung / Funktionsblock	Übertragungsverhalten		Mathematische Funktionen		Grafische Darstellungen	
	statisch	dynamisch	Zeitbereich	Frequenzbereich	Ortskurve	Bodediagramm
P-Regler (= P-Glied)  (analog: I-Regler = I-Glied)			$y(t) = K_P \cdot e(t)$	$F(j\omega) = K_P$		
PI-Regler  (PID—Block mit deaktiviertem D-Anteil)	-		$y(t) = K_P \cdot \left(e(t) + \frac{1}{T_i}\int e(t)\,dt\right)$	$F(j\omega) = K_P\left(1 + \frac{1}{j\omega T_i}\right)$	-	
PD-Regler  (PID—Block mit deaktiviertem I-Anteil)			$y(t) = K_P \cdot \left(e(t) + T_d \cdot \dot{e}(t)\right)$	$F(j\omega) = K_P(1 + j\omega T_d)$ PD-T$_1$: $F(j\omega) = K_P\left(1 + \frac{j\omega T_d}{1 + j\omega T_1}\right)$	-	
PID-Regler 	-		$y(t) = K_P \cdot \left(e(t) + \frac{1}{T_i}\int e(t)\,dt + T_d \cdot \dot{e}(t)\right)$	$F(j\omega) = K_P\left(1 + \frac{1}{j\omega T_i} + j\omega T_d\right)$ PID-T$_1$: $F(j\omega) = K_P\left(1 + \frac{1}{j\omega T_i} + \frac{j\omega T_d}{1 + j\omega T_1}\right)$	-	

Grundglieder

F	Frequenzgang (komplex)	$\Re(F)$	Realteil	$A = \lvert F \rvert$	Amplitudengang (Betrag) [dB]	$F(j\omega) = \frac{b_0 + j\omega b_1 + (j\omega)^2 b_2 + \cdots + (j\omega)^m b_m}{a_0 + j\omega a_1 + (j\omega)^2 a_2 + \cdots + (j\omega)^n a_n}$	$A_{dB} = 20 \cdot \log_{10}(A)$
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